Nelson Jeanrenaud

SOFTWARE ENGINEER | .NET & INDUSTRIAL SOFTWARE

contact@nelson-j.ch | (+41) 79 890 77 58 | Bussigny, Switzerland | LinkedIn | Portfolio

Software engineer developing .NET applications for industrial systems, including operator interfaces, PLC integration and simulation. Work covers requirements analysis, domain modelling, software architecture, implementation and testing.

EXPERIENCE

HEIG-VD - Institute of Industrial Automation | Yverdon-les-Bains, VD **Apr 2025 - Present**

R&D Collaborator

- Design the software architecture and develop control software for an automated hydroponic installation.
- Translate partner needs and operational constraints into requirements, domain models and UML designs.
- Build a Blazor operator interface with ASP.NET Core and C#, integrate Siemens and Beckhoff PLCs through OPC UA and TwinCAT ADS, and develop a WPF-based 3D digital twin.
- Coordinate technical decisions with industrial partners and automation and AI engineers; contribute to planning, automated testing and deployment.

Swiss Armed Forces **Jul-Nov 2024; Mar-Apr 2025**

Compulsory military service

Abraxas Informatik AG | Morges, VD **Aug 2023 - Oct 2024**

Software Developer

- Maintained and extended CARI, a large Java application used by Swiss road traffic offices.
- Translated business needs into functional changes and delivered them in an agile five-person development team.
- Debugged, reviewed and refactored existing code while preserving application behaviour.

HEIG-VD | Yverdon-les-Bains, VD **Aug 2023 - Oct 2024**

Teaching Assistant - Object-Oriented Programming & Cloud Computing

- Supported students in Java object-oriented modelling and cloud computing labs.

EDUCATION

HEIG-VD | Yverdon-les-Bains, VD **2023**

B.Sc. Computer Science and Communication Systems - Software Engineering

- Thesis (grade 5.6/6): Machine Learning for Making the Decision to Pit Stop in Formula 1 - data preparation, Random Forests, neural networks, and explainability.

CFPT | Geneva, GE **2020**

Federal VET Diploma in Information Technology - Software Development; Professional Maturity

TECHNICAL SKILLS

Programming languages: C#, Java, Python, SQL, JavaScript, TypeScript
Frontend & desktop: Blazor, WPF, MudBlazor, React, AngularJS
Backend frameworks: ASP.NET Core, Entity Framework Core, Spring, FastAPI
Industrial integration: Siemens PLCs (OPC UA), Beckhoff PLCs (TwinCAT ADS), REST APIs
Databases: SQLite, MySQL
Architecture & design: Object-oriented design, Clean Architecture, domain-driven design (DDD), UML
Development & testing: Git, GitLab/GitHub CI/CD, xUnit, Serilog
Deployment: Linux (ARM64), NVIDIA Jetson, PowerShell, Bash, systemd
Data & ML libraries: Pandas, NumPy, Matplotlib, scikit-learn, TensorFlow, Keras
Cloud & infrastructure: AWS, Google Cloud (GCP), Docker, Kubernetes, Terraform, Ansible

LANGUAGES

French (native), English (fluent)